HDOC DAY – 69

Algorithm

1. Start.
2. Read the number n.
3. Initialize the recursive call counter to 0.
4. Call the recursive function fibonacci() for each position from 0 to n - 1.
5. Increment the call counter whenever fibonacci() is called.
6. If n = 0, return 0.
7. If n = 1, return 1.
8. Otherwise, return fibonacci(n - 1) + fibonacci(n - 2).
9. Display each Fibonacci number.
10. Display the total number of recursive calls.
11. Stop.

C PROGRAM:

```
#include <stdio.h>

int count = 0;

int fibonacci(int n)
{
   count++;

   if (n == 0)
      return 0;

   if (n == 1)
      return 1;
```

```
    return fibonacci(n - 1) + fibonacci(n - 2);
}

int main()
{
    int n, i;

    printf("Enter the number of Fibonacci numbers: ");
    scanf("%d", &n);

    printf("Fibonacci series: ");

    for (i = 0; i < n; i++)
        printf("%d ", fibonacci(i));

    printf("\nNumber of recursive calls: %d\n", count);

    return 0;
}
```

TEST CASE TABLE:

Test Case	n	Fibonacci Series	Recursive Calls
1	1	0	1
2	2	0 1	2
3	3	0 1 1	5
4	5	0 1 1 2 3	19
5	6	0 1 1 2 3 5	34